Riem. curvature:

As a $\binom{1}{3}$ tensor,

$$R(X,Y)Z := \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z$$

Think of $R(\cdot,\cdot)$ as a map $\mathfrak{X}M \to \mathfrak{X}M$

As a $\binom{0}{4}$ tensor, define

$$Rm(X,Y,Z,W) := \langle R(X,Y)Z, W \rangle$$

In coords, $R(\partial_i, \partial_j)\partial_k = R^l_{ijk}\,\partial_l$

$$Rm(\partial_i, \partial_j, \partial_k, \partial_m) = R_{ijkm} = R^l_{ijk}\, g_{lm}$$

Symmetries of Rm

1) by def. $Rm(X,Y,\cdot,\cdot) = -Rm(Y,X,\cdot,\cdot)$

2) ∇ is compatible w/ g $\Rightarrow$ $Rm(\cdot,\cdot,Z,W) = -Rm(\cdot,\cdot,W,Z)$

3) ∇ is symmetric $\longrightarrow$ $Rm(X,Y,Z,W) = Rm(Z,W,X,Y)$

This cuts the number of independent functions in Rm from $n^4 \longrightarrow \dfrac{n^2(n^2-1)}{12}$

Bianchi identities: From above we can deduce a cyclic sum

$$Rm(X,Y,Z,W) + Rm(Y,Z,X,W) + Rm(Z,X,Y,W) = 0$$

There is also a "differential" Bianchi identity:

$$\nabla_Z R(X,Y) + \nabla_X R(Y,Z) + \nabla_Y R(Z,X) = 0$$

Average curvatures

Ricci Rc is a symmetric 2-tensor defined by

$$Rc(Y,Z) = \operatorname{tr} Rm(\cdot, Y, Z, \cdot)$$

meaning $\sum_{i=1}^{n} Rm(E_i, Y, Z, E_i)$ where E_i is orthonormal

in coord.'s $\quad R_{jk} = R_{ijkl}\, g^{il}$

$$= R^{i}_{ijk}$$

Ricci curvature is much more likely to appear in applications than Rm

example: Einstein equations in GR

Scalar curvature

$$S = \operatorname{tr} Rc(\cdot,\cdot) = g^{jk} R_{jk}$$

(in dim. 2, $S = 2K$)

The other important form is **sectional curvature**

= Gauss curvature of submanifolds of dim 2.

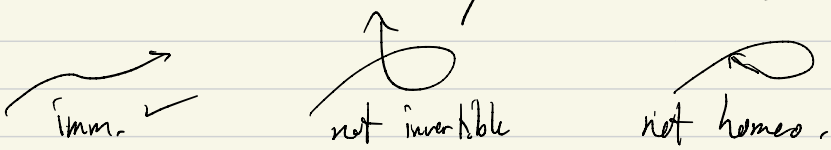
Riemannian submanifolds

Manifold maps $f : M \longrightarrow \tilde{M}$ (smooth)

immersion : df_p is injective $T_pM \to T_{f(p)}\tilde{M}$ for all p

embedding : immersion such that f is a homeomorphism $M \to f(M)$

ex: A regular curve is an immersion $I \to M$, but not necessarily an embedding



On top of these, isometry means $g = f^* \tilde{g}$

(A unit speed curve is an isometric immersion)

def: A Riemannian submanifold of $\tilde{M}$ is the image of an isometric embedding $M \to \tilde{M}$

We normally identify M with its image and write $M \subset \tilde{M}$

example: A regular surface in $\mathbb{R}^3$ was defined as a Riem. submfld.

Normal bundle ($M \subset \tilde{M}$ Riem. submfld)

For $p \in M$, T_pM is a subspace of $T_p\tilde{M}$

The normal vector space is $N_pM := (T_pM)^\perp$ with respect to $\tilde{g}$

$$\Rightarrow T_p\tilde{M} = T_pM \oplus N_pM$$

$$X = X^T + X^\perp$$

normal bundle $NM := \bigcup_p N_pM$

$\mathcal{N}(M) = \{\text{normal vector fields}\}$

For $X, Y \in \mathfrak{X}M$, define the second fundamental form

$$\mathrm{II}(X,Y) := (\tilde{\nabla}_X Y)^\perp$$

(the def. requires smooth extensions to $\tilde{M}$)

Lemma (1) II is well-defined independently of extensions

(2) II is bilinear over $C^\infty(M)$

(3) II is symmetric

Pro = For $p \in M$, $\tilde{\nabla}_X Y|_p$ depends only on X_p and

$$\tilde{\nabla}_{fX} Y = f \tilde{\nabla}_X Y \quad \text{for } f \in C^\infty(\tilde{M})$$

For the Y variable it suffices to prove symmetry

$$\tilde{\nabla}_X Y - \tilde{\nabla}_Y X = [X,Y] \qquad \text{(assuming extensions)}$$

At points of M, X and Y are tangent
$\Rightarrow [X,Y]$ is also tangent

$$[X,Y]^\perp|_{T_pM} = 0$$

For surfaces in $\mathbb{R}^3$, we can decompose derivatives of a vector field into tangential and normal components

The analog here is:

Thm (Gauss formula)

For $X, Y \in \mathfrak{X}(M)$, with arbitrary extension to $\tilde{M}$,

$$\tilde{\nabla}_X Y|_M = \nabla_X Y + \mathrm{II}(X,Y)$$

Proof:

Define a connection ∇^T on M by

$$\nabla^T_X Y := (\tilde{\nabla}_X Y)^T \qquad \text{(projection } T_p\tilde{M} \to T_pM\text{)}$$

(This is how we defined covariant derivatives on surfaces.)

Check: ∇^T is compatible with g on M

example: $X\tilde{g}(Y,Z) = \tilde{g}(\tilde{\nabla}_X Y, Z) + \tilde{g}(Y, \tilde{\nabla}_X Z)$

on M, decompose $\tilde{\nabla}_X Y = (\nabla_X Y)^T + (\nabla_X Y)^\perp$

$$\Rightarrow X\tilde{g}(Y,Z)\big|_M = g(\nabla^T_X Y, Z) + g(Y, \nabla^T_X Z) \quad \checkmark$$

uniqueness of the Riem. connection $\Rightarrow \nabla^T = \nabla$

Since $\mathrm{II}(X,Y) = (\tilde{\nabla}_X Y)^\perp$, this proves the formula